

# RESULTS: KL REGULARIZATION AGAINST AN EMA REFERENCE

BRIAN BARTOLDSON

## 1. SETUP

We train Qwen3-4B-Instruct-2507 with TBA on Countdown using asynchronous rollouts ( $\Delta = 10$ ),  $\alpha = 0.9$  for the EMA reference,  $\beta = 0.005$ ,  $\text{lr} = 10^{-6}$ ,  $K = 16$  samples per problem, sequence length 4096, 720 training steps, evaluation every 20 steps.

Per-token KL.. Two choices for the per-token KL contribution to the advantage:

- *Exact KL*:  $\log T_n(y) - \log E_n(y)$ , where  $E_n$  is the EMA reference policy.
- *Approx KL*:  $\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$ , the first-order surrogate derived in `main.tex` (eq. 2.10). This avoids a forward pass through  $E_n$  at training time.

Centering. For each rollout  $i$ , the per-token KL is summed over masked tokens to give a per-rollout KL term  $K^{(i)} = \sum_t \text{kl\_per\_token}(y_t^{(i)})$ . The advantage is then adjusted by

$$(1) \quad \text{advantage}^{(i)} \leftarrow \text{advantage}^{(i)} - \beta \cdot (K^{(i)} - \bar{K}),$$

where  $\bar{K}$  is a centering term. Two choices:

- *K-centered*:  $\bar{K} = \frac{1}{K} \sum_{j=1}^K K^{(j)}$ , the within-group mean of  $\{K^{(j)}\}_{j=1}^K$  over the  $K$  rollouts sharing a problem.
- *No centering*:  $\bar{K} \equiv 0$ , so  $\text{advantage}^{(i)} \leftarrow \text{advantage}^{(i)} - \beta K^{(i)}$  directly.

## 2. 2 × 2 MATRIX: EMA REFERENCE

Crossing the two per-token KL choices with the two centering choices yields four cells:

| Cell                 | kl_per_token( $y$ )                                                    | Centering term $\bar{K}$           |
|----------------------|------------------------------------------------------------------------|------------------------------------|
| Exact, K-centered    | $\log T_n(y) - \log E_n(y)$                                            | $\frac{1}{K} \sum_{j=1}^K K^{(j)}$ |
| Exact, no centering  | $\log T_n(y) - \log E_n(y)$                                            | 0                                  |
| Approx, K-centered   | $\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$ | $\frac{1}{K} \sum_{j=1}^K K^{(j)}$ |
| Approx, no centering | $\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$ | 0                                  |

(In the K-centered rows,  $K^{(j)}$  is built from the same row’s kl\_per\_token.)

| Cell                 | Peak eval (step)   | Final eval |
|----------------------|--------------------|------------|
| Exact, K-centered    | 0.789 (240)        | 0.656      |
| Exact, no centering  | <b>0.827</b> (620) | 0.821      |
| Approx, K-centered   | 0.794 (380)        | 0.548      |
| Approx, no centering | <b>0.831</b> (520) | 0.819      |

The headline observations:

- (1) Within either KL choice, removing the centering term strictly helps.
- (2) The approx-KL row tracks the exact-KL row closely: peaks differ by 0.005 and finals by 0.002 in the no-centering pair. This is the empirical justification for substituting the first-order surrogate (no extra forward) for the exact EMA KL.
- (3) The K-centered cells exhibit the same kind of late-run instability under both per-token KL choices, suggesting the failure mode is centering-related rather than KL-formula-related.

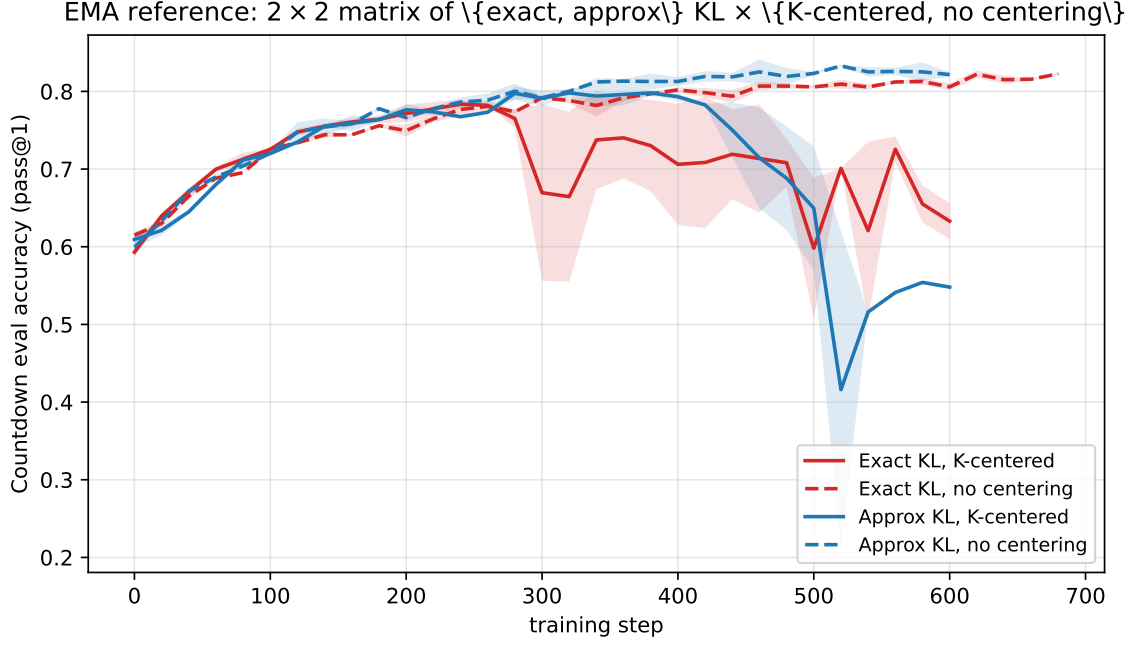


FIGURE 1. Eval accuracy on Countdown for the four cells of the EMA-reference matrix. Red: exact KL. Blue: approx KL. Solid: K-centered. Dashed: no centering. The two no-centering cells (dashed) plateau above 0.80 and peak above 0.82; the two K-centered cells (solid) peak earlier and degrade after step  $\approx 300$ .

### 3. RESET REFERENCE (BASELINE)

Before the EMA work, the same form of objective was tested with a periodic-reset reference policy  $R$  (hard copy from the training policy every 50 rollout steps). The advantage update (1) is unchanged, but per-token KL is computed against  $R$  instead of  $E_n$ , and the source for  $\log T(\cdot)$  becomes configurable. Let  $T_n$  denote the live training-policy forward at the current step and  $T_{n-\Delta}$  the inference snapshot used during rollout generation. The per-sample per-token KL is

$$(2) \quad \log T_{\text{src}_1}(y) - \log R(y),$$

and the K-group mean term substitutes a (potentially different) source  $T_{\text{src}_2}$ :

$$(3) \quad \bar{K} = \frac{1}{K} \sum_{i=1}^K \sum_t \left( \log T_{\text{src}_2}(y_t^{(i)}) - \log R(y_t^{(i)}) \right),$$

where each  $\text{src}_j \in \{T_n, T_{n-\Delta}\}$ . We test three combinations, each with importance sampling on or off, all with K-centering:

| Variant (short) | Per-sample $\log T_{\text{src}_1}$ | K-group mean $\log T_{\text{src}_2}$     |
|-----------------|------------------------------------|------------------------------------------|
| $T/I$           | $\log T_n$ (live train forward)    | $\log T_{n-\Delta}$ (inference snapshot) |
| $T/T$           | $\log T_n$                         | $\log T_n$                               |
| $I/I$           | $\log T_{n-\Delta}$                | $\log T_{n-\Delta}$                      |

The first slot is the per-sample  $\log T$ ; the second is the K-mean  $\log T$ . Historically these mapped to “klMeanInference”, “klMeanTrain”, and “klAllInf” respectively.

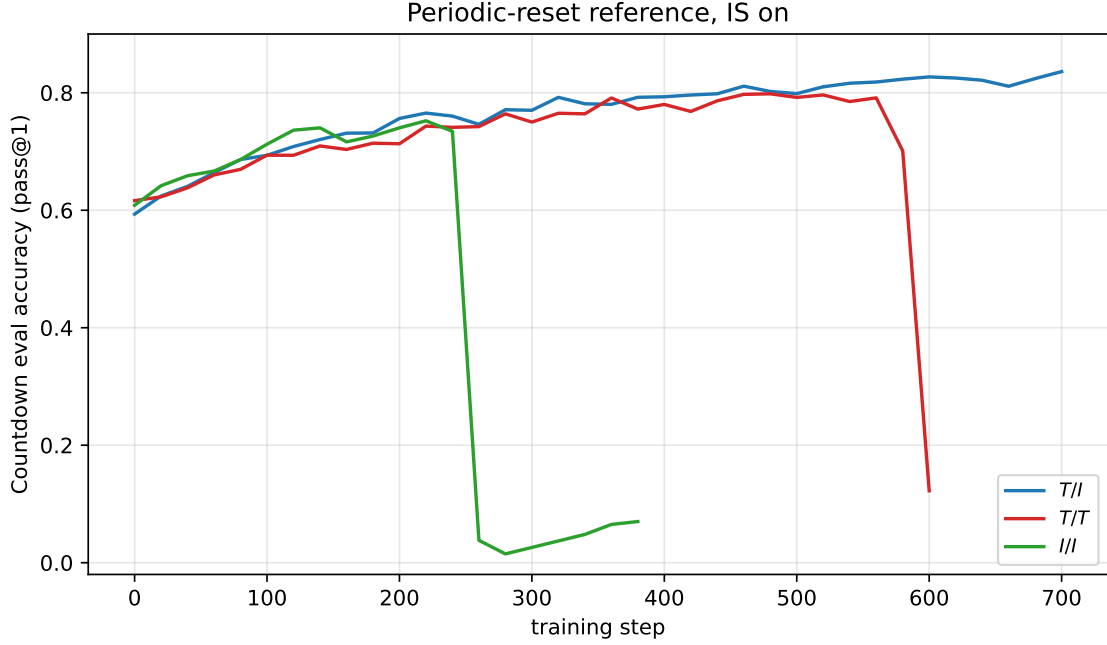


FIGURE 2. Eval accuracy on Countdown with periodic-reset reference and importance sampling on.  $T/T$  (red) collapses late in training;  $I/I$  (green) is unstable; only  $T/I$  (blue) trains stably to high accuracy.

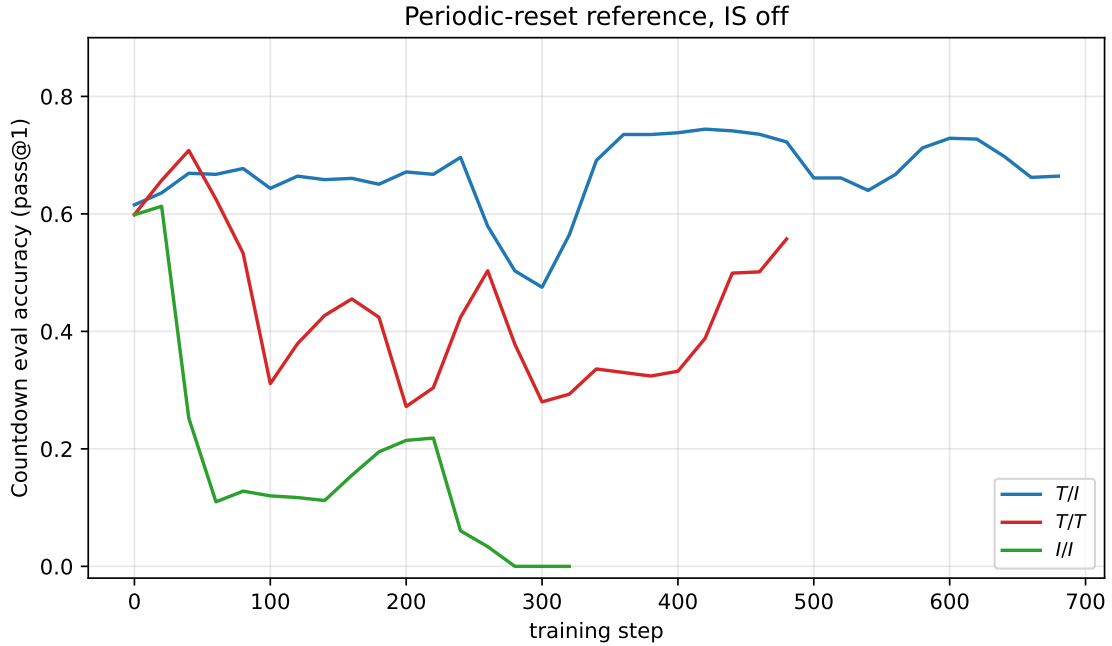


FIGURE 3. Eval accuracy on Countdown with periodic-reset reference and importance sampling off. The same ordering holds —  $T/I$  best,  $T/T$  collapses,  $I/I$  gets stuck near zero — and all three are weaker than their IS-on counterparts.

| Run           | Peak eval (step) | Final eval |
|---------------|------------------|------------|
| $T/I$         | 0.836 (700)      | 0.836      |
| $T/T$         | 0.798 (480)      | 0.122      |
| $I/I$         | 0.752 (220)      | 0.070      |
| $T/I$ , no IS | 0.744 (680)      | 0.664      |
| $T/T$ , no IS | 0.708 (40)       | 0.557      |
| $I/I$ , no IS | 0.613 (20)       | 0.000      |

The  $T/T$  collapse under reset reference mirrors the K-centered instability in the EMA matrix: in both cases, the per-sample and K-mean KL terms are computed from the *same* drifting source  $T_n$ , so the centering subtraction tracks the drift rather than canceling it.

#### 4. APPROXIMATION ACCURACY

We track the per-token error of the first-order surrogate against the exact EMA KL inside `tba_loss`: **MAE** (mean absolute error over masked tokens) and **bias** (signed mean). Figure 4 compares the two no-centering cells (where the loss path differs but both have the same telemetry).

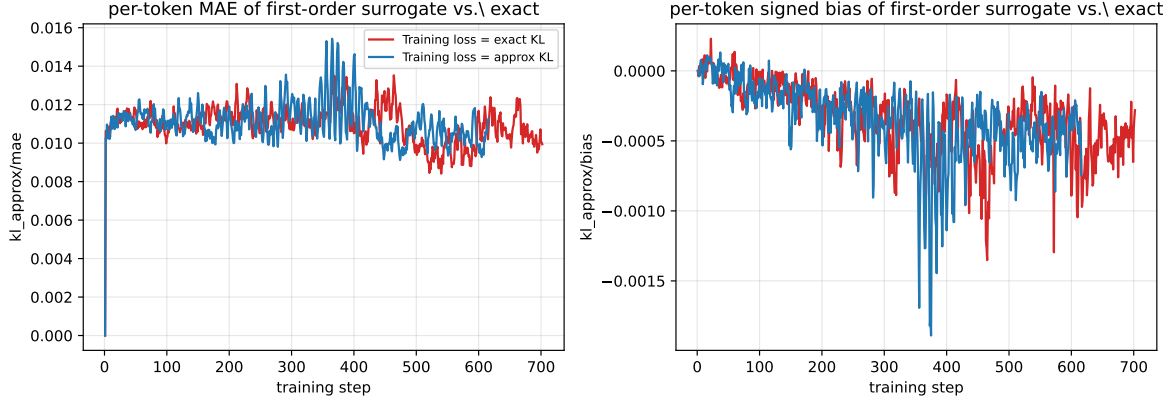


FIGURE 4. Per-token MAE (left) and signed bias (right) of the first-order surrogate  $\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$  against the exact  $\log T_n(y) - \log E_n(y)$ , averaged over masked tokens within each rollout. Both runs are no-centering EMA-reference cells; they differ only in which formula enters the loss (red: exact KL; blue: approx KL).

MAE stays in the low  $10^{-2}$  band throughout training, and the signed bias is at most a few parts in  $10^3$ . The approximation is empirically faithful, regardless of whether it is used in the loss.

Across the  $\alpha$ -sweep. The same metric is logged for every  $\alpha$  run. The intrinsic MAE scales with  $c_\alpha$ , since the exact KL grows like  $c_\alpha \cdot (\log T_n - \log T_{n-\Delta})$ . We therefore also track a rescaled quantity,  $(c_{0.9}/c_\alpha) \text{KL}_{\text{exact}} - c_{0.9} \text{diff}$ , whose magnitude should be roughly  $\alpha$ -invariant if the first-order surrogate has the same fractional quality across  $\alpha$ .

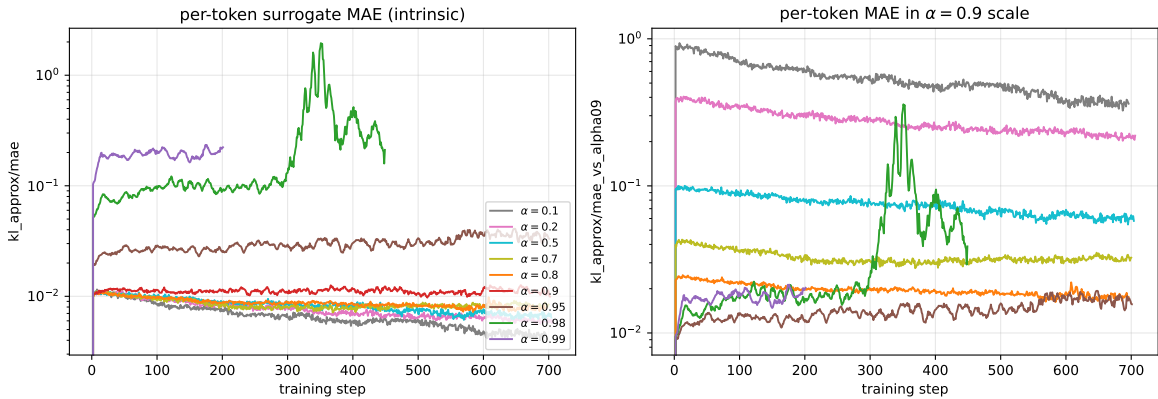


FIGURE 5. Per-token surrogate MAE across the  $\alpha$ -sweep. Left: intrinsic MAE (scales with  $c_\alpha$ ). Right: MAE after rescaling to the  $\alpha = 0.9$  surrogate; comparable across  $\alpha$  values.

## 5. ABLATIONS

We probe three knobs in the no-centering / K-centered EMA cells. Some of the runs cited here are still in progress, so trend lines for the newest variants may be short.

**5.1.  $\beta$  ablation on the K-centered cell (A).** Cell A (EMA exact + klMeanTrain centering) showed a late-run collapse around step  $\sim 300$  at the baseline  $\beta = 0.005$ . Does scaling the KL penalty up rescue it?

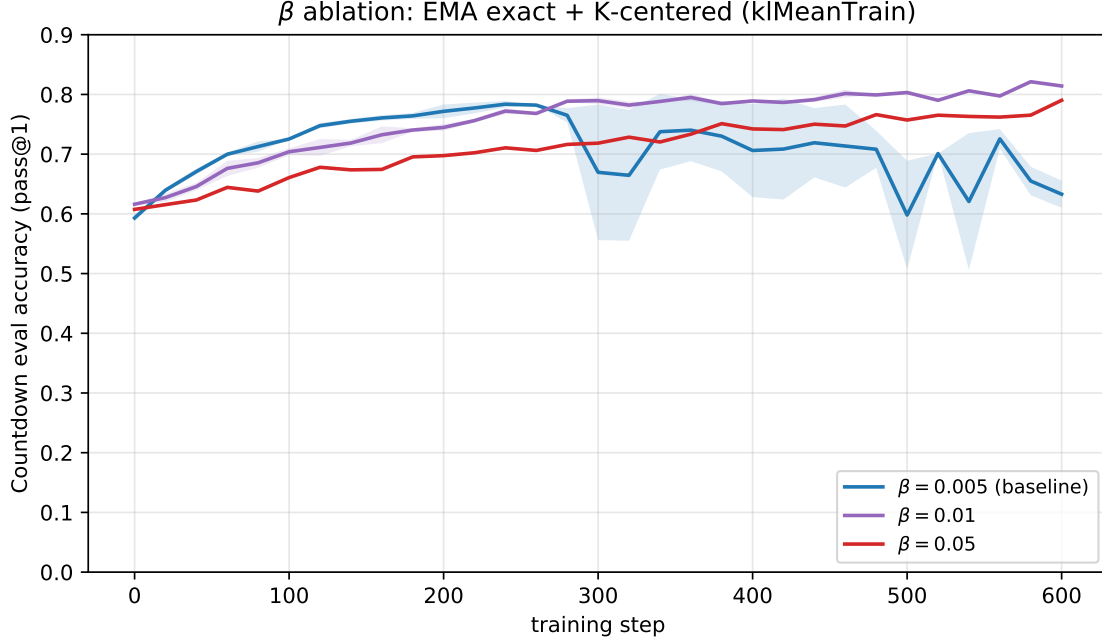


FIGURE 6.  $\beta$  ablation on K-centered exact KL (cell A). Larger  $\beta$  trades off learning speed against the K-centered instability.

**5.2. Importance sampling ablation on no-centering.** The no-centering cells B and D (exact and approx) were the top-performing matrix entries at  $\beta = 0.005$  with IS on. Turning IS off keeps the same conclusion that no centering is best, but at lower absolute accuracy and with late-run instability.

**5.3.  $\alpha$  ablation on no-centering exact.** Sweep the EMA decay  $\alpha \in \{0.1, 0.2, 0.5, 0.7, 0.8, 0.9, 0.95, 0.98, 0.99\}$  while holding the effective regularization strength constant ( $\beta \cdot c_\alpha = 0.0045$ , where  $c_\alpha = \alpha/(\Delta(1 - \alpha))$ ).

**5.4.  $\beta$  ablation on no-centering, IS off (B-noIS).** The no-centering exact cell with IS disabled also shows late-run instability at the baseline  $\beta = 0.005$ . Does scaling  $\beta$  up help here too?

**5.5.  $\beta$  ablation on reset reference,  $T/T$ .** The reset-reference  $T/T$  variant collapsed late at the baseline  $\beta = 0.005$ . Does scaling  $\beta$  up rescue the collapse, as it might for the analogous EMA-reference K-centered cell?

**5.6.  $\beta$  ablation on reset reference,  $I/I$ .** The  $I/I$  variant uses the inference snapshot for both per-sample and K-mean KL. Since per-sample KL uses inference logprobs, we modified the reference reset to copy from the inference snapshot rather than the live train model (so  $R$  consistently lags  $T_{\text{inf}}$ ). We then probe whether raising  $\beta$  helps the  $I/I$  cell.

**5.7.  $\beta$  ablation on no-centering, approx KL (cell D).** We also sweep  $\beta$  for the approx-KL no-centering cell to test the regularization sensitivity of the approx-KL path (cell D,  $\alpha = 0.9$ ).

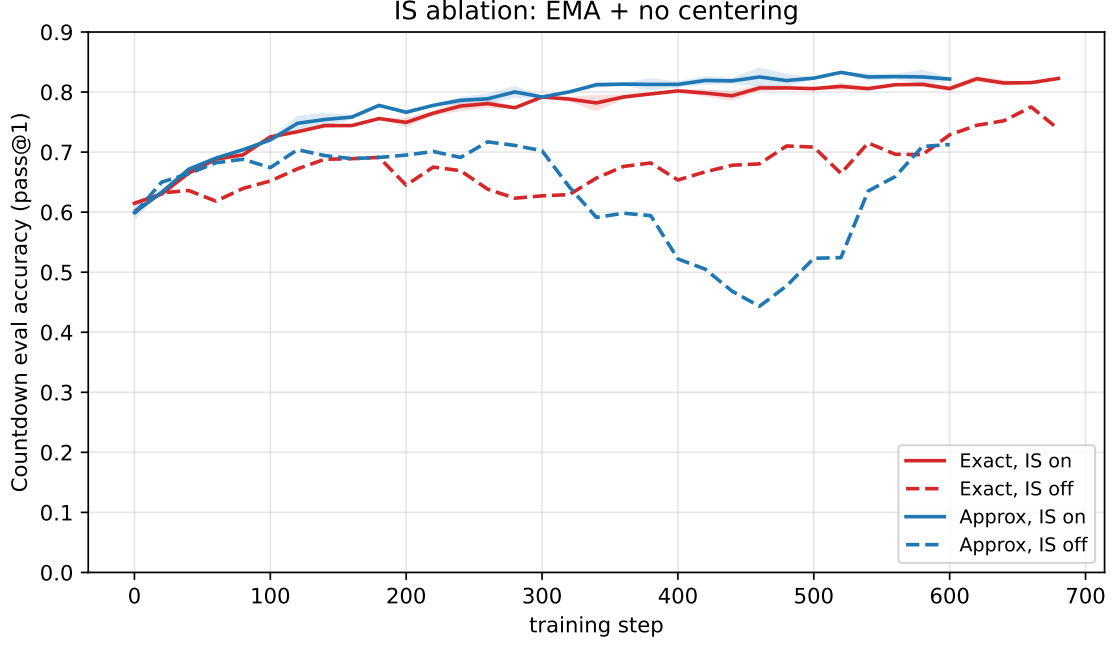


FIGURE 7. IS ablation on no-centering cells. Solid: IS on. Dashed: IS off. Red: exact KL (cell B). Blue: approx KL (cell D).

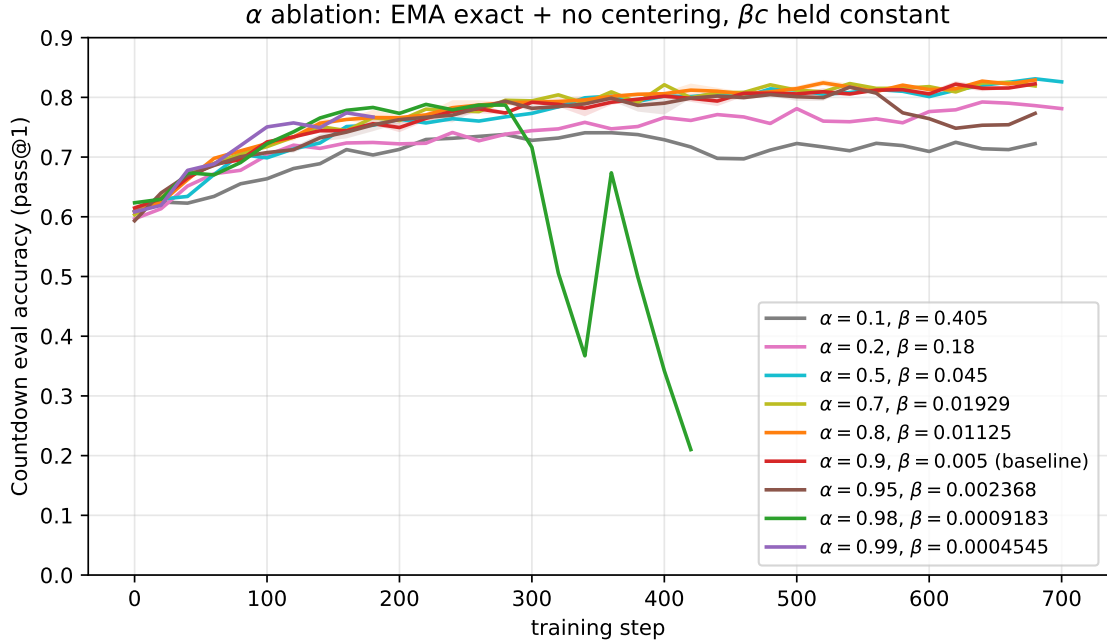
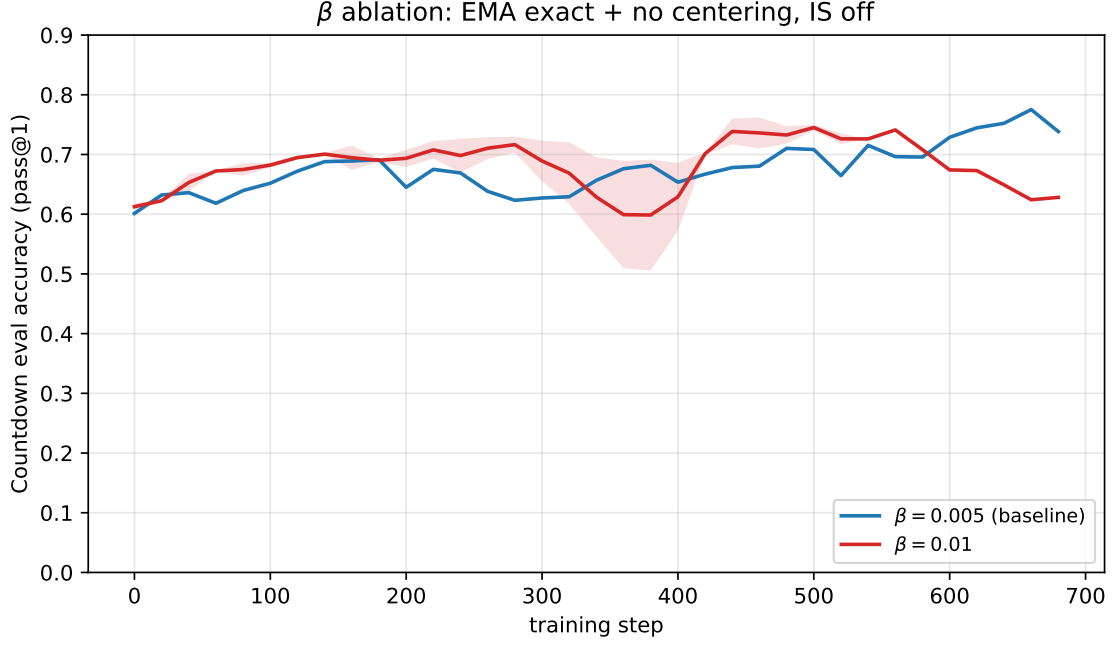
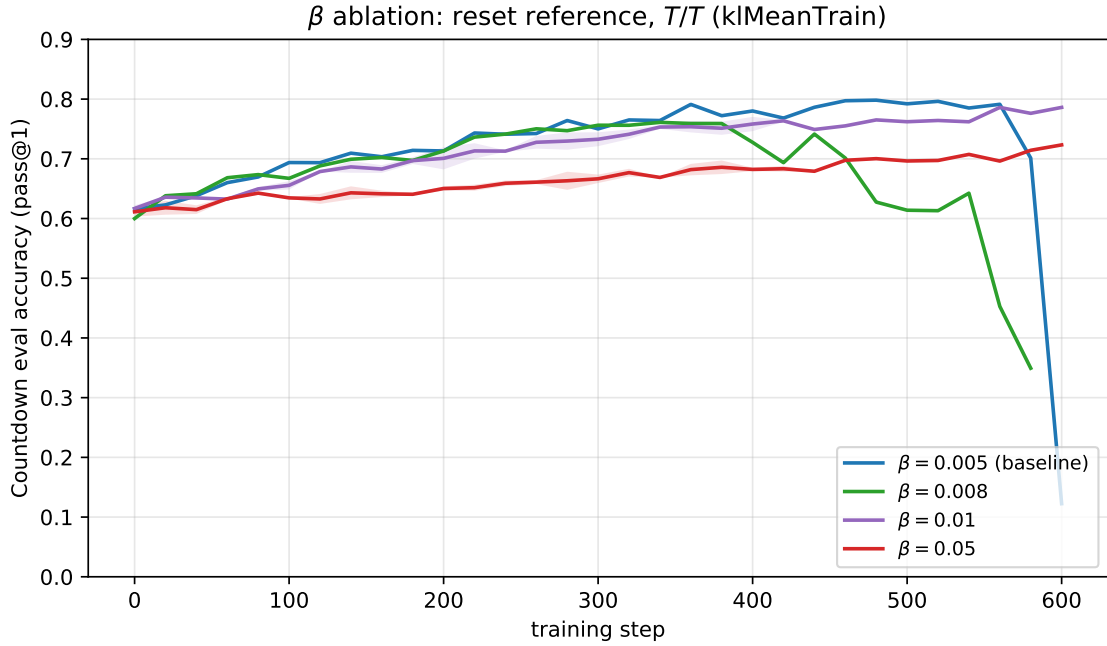


FIGURE 8.  $\alpha$  ablation on no-centering exact KL with  $\beta$  scaled to keep  $\beta \cdot c_\alpha$  constant. Higher  $\alpha$  (slower EMA) becomes unstable:  $\alpha = 0.99$  collapses around step  $\sim 200$ ,  $\alpha = 0.98$  partially collapses around step  $\sim 300-400$ ;  $\alpha \in \{0.8, 0.95\}$  match or beat the  $\alpha = 0.9$  baseline.

**5.8. Application: pipeline / heterogeneous-async RL.** The  $\beta \cdot c$  invariance suggests a per-rollout calibration in async pipelines where rollouts originate from inference snapshots of different ages. Concretely, if rollout  $i$  was generated from a snapshot  $\Delta_i$  steps behind the current trainer, its implicit  $c_i = \alpha / (\Delta_i(1 - \alpha))$  varies. Scaling the KL coefficient as  $\beta_i \propto 1/c_i$  keeps the

FIGURE 9.  $\beta$  ablation on no-centering exact KL with IS off (B-noIS family).FIGURE 10.  $\beta$  ablation on the reset-reference  $T/T$  cell (klMeanTrain). Larger  $\beta$  prevents the late-run instability seen at  $\beta = 0.005$  (no collapse observed through step  $\sim 580$  at  $\beta = 0.01$  or  $\beta = 0.05$ ).

regularization strength fixed across rollouts—the analogue of importance-sampling reweighting for the policy gradient, but for the KL term. This is the strongest application direction we see for these findings; we leave the experimental study for future work.

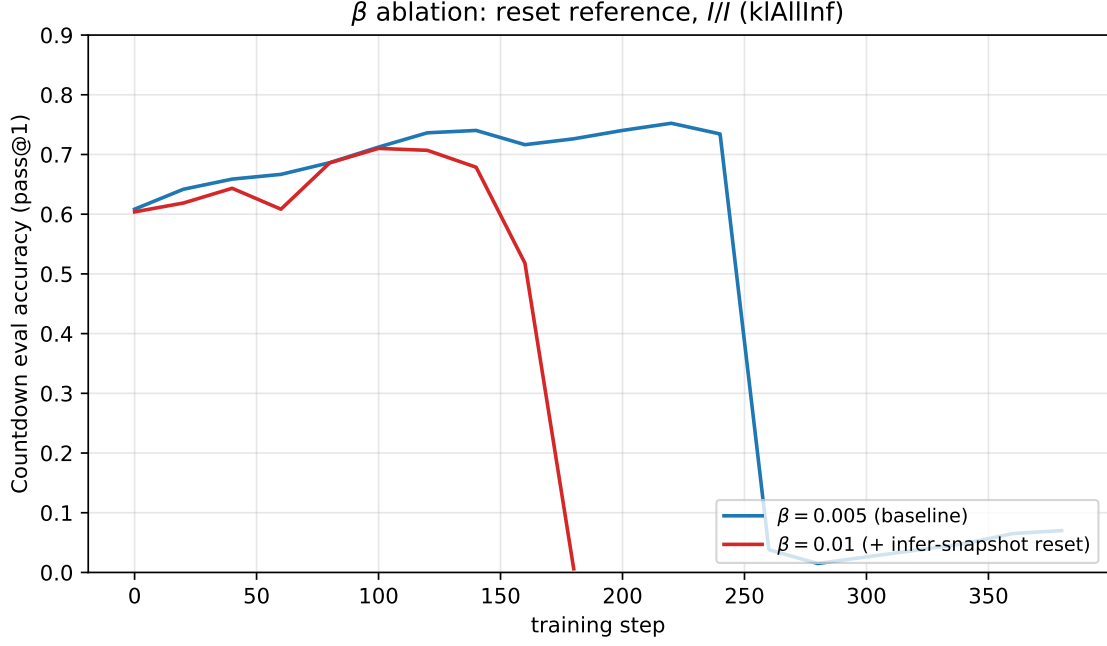


FIGURE 11.  $\beta$  ablation on the reset-reference  $I/I$  cell (klAllInf). The  $\beta = 0.01$  variant also uses the inference-snapshot reset fix.

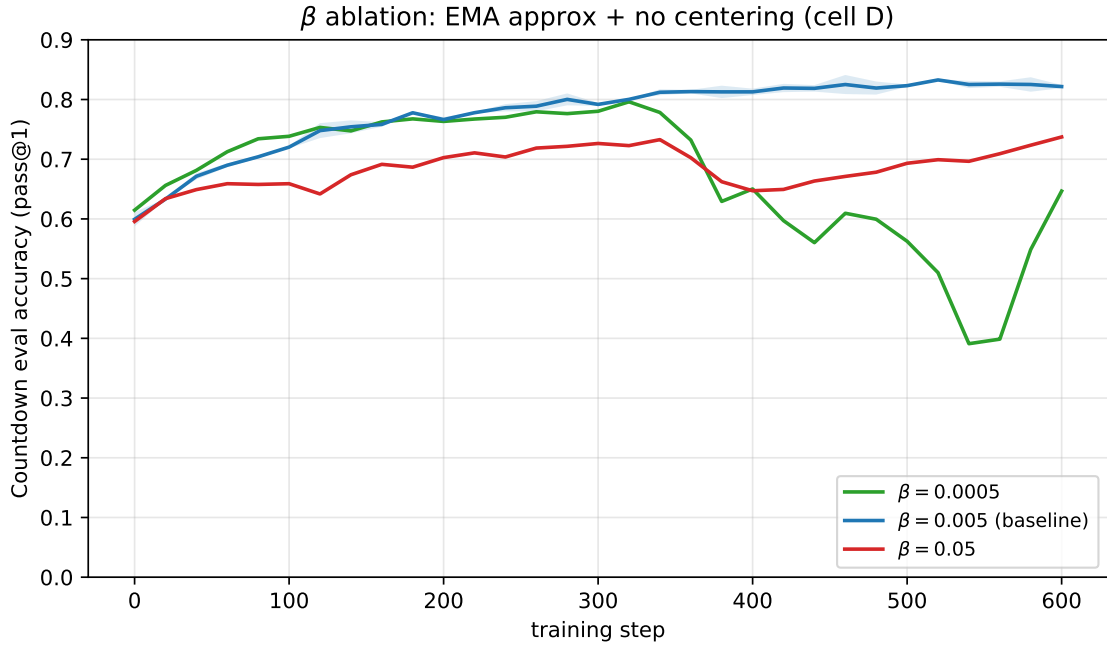


FIGURE 12.  $\beta$  ablation on no-centering approx KL (cell D). Baseline  $\beta = 0.005$  vs.  $\beta = 0.05$  ( $10\times$ ) and  $\beta = 0.0005$  ( $0.1\times$ ). The cell is robust to weakening  $\beta$  (no early-run impact at  $\beta = 0.0005$ ) but slower learning at  $\beta = 0.05$ .

## 6. CONCLUSION

Removing the K-group centering is the dominant lever: both EMA exact and EMA approx benefit. With centering off, the cheap first-order surrogate matches the exact EMA KL within 0.005 on peak eval and 0.002 on final eval, so it is a viable drop-in replacement that saves a forward pass through the reference model.